ScienceQA answer prediction

Early and late fusion with frozen encoders

Do image features improve a text model?

Official splits • CPU execution • Locked evaluation

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Data and fusion methods

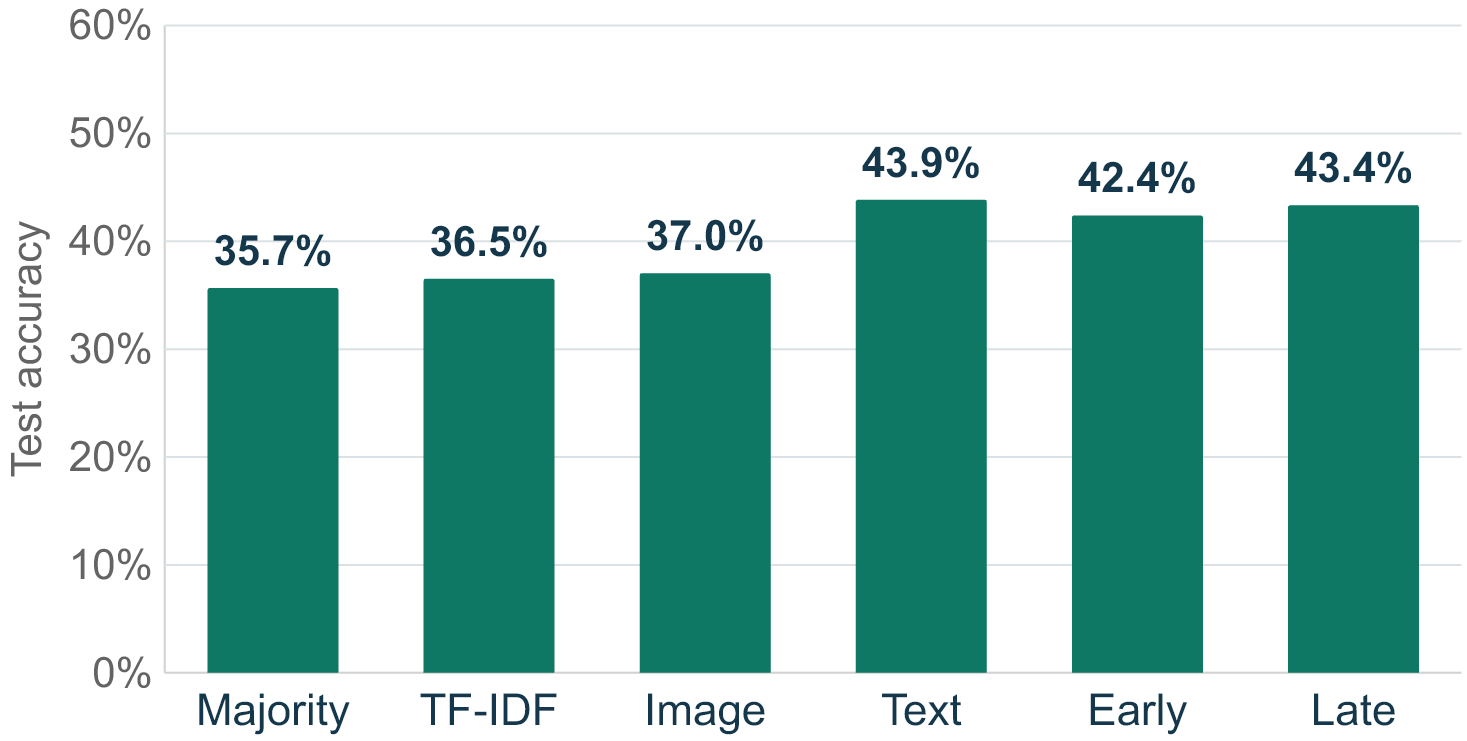
6,108 train / 2,053 validation / 1,979 test

Image-bearing questions with 2–4 choices. No lectures or solutions as inputs.

Model	Representation	Decision
Text only	MiniLM · 1,920 features	128-unit MLP
Image only	ResNet-18 · 512 features	128-unit MLP
Early fusion	Concatenate · 2,432 features	128-unit MLP
Late fusion	Separate probabilities	0.70 text + 0.30 image

Train-only scaling. Legal-choice masks. Validation-NLL selection before testing.

Fusion did not improve answer accuracy



Probability quality

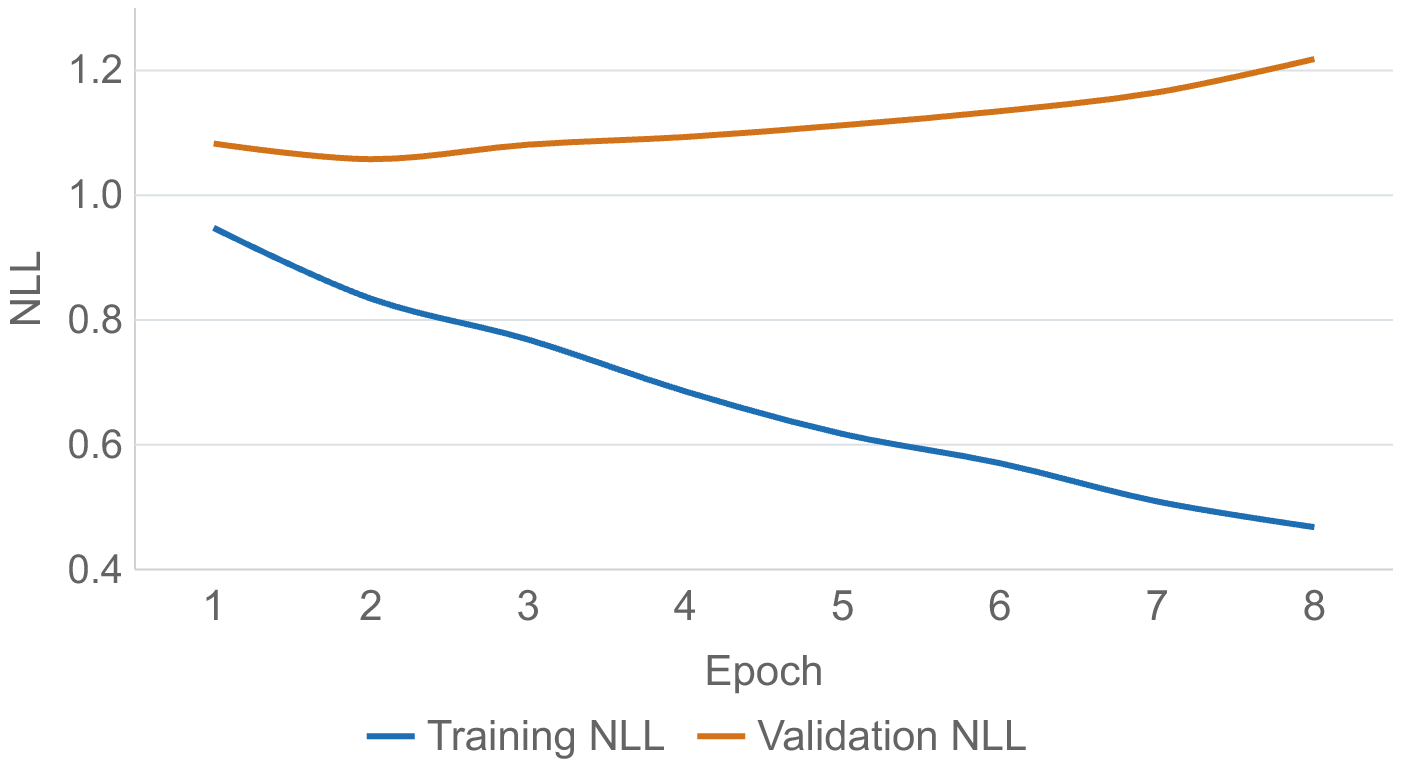
Metric	Text	Late
NLL	1.071	1.055
Brier	0.638	0.629

Probability error scores
Lower is better.

Primary seed 42 · n = 1,979 · Uniform guess expectation 35.79%

95% difference intervals include zero; no demonstrated accuracy gain.

Overfitting and limited image contribution



Epoch 2 selected

Early fusion
Train 63.97%
Validation 42.72%

Image shuffling
42.40% → 41.74%

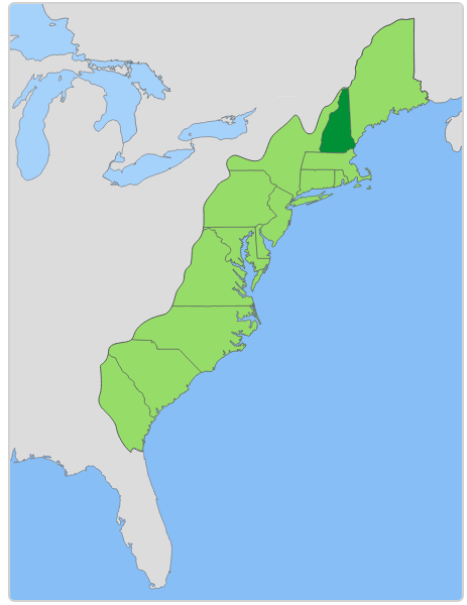
555 test images also occur in training. Unseen-image text accuracy: 39.26% (n = 1,424).

Two concrete failure cases



Lost experimental context

253 tokens → 128 tokens
Wind-tunnel conditions omitted
Prediction: damage detection
Correct: parachute steadiness



Incorrect spatial recognition

Prediction: Rhode Island
Correct: New Hampshire

First two early-fusion errors in test order. Images: ScienceQA, CC BY-SA 4.0.

Conclusion and practical limits

Text alone: 43.86% accuracy

Late fusion improves probability scores, not answer accuracy.

Efficiency

~11.2 min feature extraction
<3 s per primary neural head
CPU · 4 threads

Limits and next steps

Repeated templates and subject imbalance
Check longer context and OCR on new data
Human review for educational use

AI assistance disclosed in report. Code and measured results are documented.